

Aligning Financial Asset Recommendations with Investor Risk Profiles via Profile Coherence

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Abstract

Financial asset recommendation (FAR) literature optimises ranking quality and realised return, but no published metric exposes whether recommendations are profile-aligned in the regulatory sense that MiFID II suitability requires. We introduce Profile Coherence at k (PC@ k), scoring the share of a top- k list whose asset risk class lies within ± 1 band of the customer’s declared MiFID II risk band on a four-point ordinal scale. Applying PC@ k to FAR-Trans [Sanz-Cruzado *et al.*, 2024], we find that 18.6% of 228,241 scoreable Buys violate the declared band by two or more bands, with a sharply bimodal per-customer self-discordance distribution. Yet a transaction-level Ordinary Least Squares (OLS) regression on 208,029 priced Buys, with cluster-robust standard errors on `customerID`, finds coherent Buys earn +2.94 percentage points (pp) higher 6-month realised return than discordant ones, conditional on volatility, segment, and year. We then propose a LightGCN extension that stratifies by band and adds a profile-coherent margin loss atop Bayesian Personalised Ranking (BPR), lifting aggregate PC@10 from 0.794 to 0.918 across all 69 temporal evaluation splits, at negligible aggregate return on investment cost.

1 Introduction

Financial asset recommendation (FAR) sits at the intersection of personalised ranking and regulatory obligation. The European Markets in Financial Instruments Directive (MiFID II) requires firms to assess suitability before recommending a product: alignment between the product’s risk profile and the customer’s declared risk tolerance, investment horizon, and loss capacity. A recommender that maximises predicted return without conditioning on the customer’s risk band can be an excellent ranker on standard benchmarks but a regulatory liability.

FAR-Trans [Sanz-Cruzado *et al.*, 2024] is the first open FAR benchmark with both customer profile metadata and a multi-year transaction history: 228,913 Buys across 806 assets and 29,090 retail customers, with MiFID II risk bands per customer and metadata sufficient to derive each asset’s

risk class. Of these, 228,241 Buys are *scoreable*: both the customer and the asset have an assigned risk band, enabling pairwise discordance measurement. The accompanying paper benchmarks LightGCN (a graph-based collaborative filter that propagates embeddings over the user-item bipartite graph), Random Forest, and others on normalised Discounted Cumulative Gain (nDCG) and Return on Investment (ROI), each evaluated on the top-10 ranked items. We additionally report Recall alongside these as a ranking sanity check. None of those metrics measures profile alignment, and none of the published baselines conditions on the declared risk band.

This paper introduces a profile-alignment lens for FAR and applies it in four steps: *diagnose* the prevalence of discordance, *test* whether it carries an economic penalty, *audit* where existing baselines sit on the new axis, and *remedy* the gap with a profile-coherent training objective. These map onto four research questions:

- **RQ1 (Diagnose):** How prevalent is profile-discordance in observed FAR-Trans Buy transactions, and is it a stable customer trait or transaction-level noise?
- **RQ2 (Test the economic objection):** Do profile-discordant Buys earn lower realised six-month return than coherent ones, once asset volatility, customer segment, and year are controlled for?
- **RQ3 (Audit existing baselines):** At their best published settings, do the chosen FAR-Trans baseline models lift profile coherence above the band-conditional random baseline $\pi(b)$ (the expected coherence of a uniformly random recommender) for every declared band?
- **RQ4 (Remedy):** Can a stratified, profile-coherent LightGCN improve profile coherence over the global baseline, and at what cost to nDCG, Recall, and ROI?

Contributions. This paper makes two contributions. First, we introduce Profile Coherence at top- k ranked items (PC@ k), a coherence metric with a band-conditional random baseline $\pi(b)$ and scale-invariant lift normalisation. Second, we propose a stratified, profile-coherent LightGCN with a margin loss atop Bayesian Personalised Ranking (BPR, the standard pairwise loss for LightGCN), together with an ablation that isolates the coherence-loss effect from the stratified architecture.

2 Related Work

2.1 FAR benchmarks and the two-axis evaluation status quo

FAR-Trans [Sanz-Cruzado *et al.*, 2024] is the first public FAR benchmark with both real retail-investor transactions and asset pricing, and it establishes the evaluation standard for the field: 11 baselines are reported on nDCG@10 for next-purchase ranking quality and ROI@10 for the realised return. The subsequent study by Sanz-Cruzado *et al.* [2026] shows that these two axes are not aligned on FAR-Trans: transaction-based and profitability-based metrics are negatively correlated, and models that win one lose the other. However, neither axis measures whether a recommendation aligns with the customer’s declared MiFID II risk band, which is the gap PC@ k addresses.

2.2 Profile-aware FAR architectures

Ghiye *et al.* [2024] extend LightGCN with a causal graph convolution and a sliding-window training scheme for credit-bond recommendation, capturing temporal drift in user interests via embedding updates restricted to recent interactions. Their result supports the broader point that LightGCN can be modified to carry domain-specific structure beyond plain Bayesian Personalised Ranking (BPR), and motivates the choice of LightGCN as the backbone for our stratified, profile-coherent extension.

2.3 Risk-aware FAR via post-hoc re-ranking

Sakurai *et al.* [2026] propose a risk-aware utility re-ranking (RURA) procedure on FAR-Trans: a mean-variance utility conditioned on the customer’s MiFID risk tolerance is applied to a base ranker’s top list, lifting ROI@10 while preserving nDCG@10. RURA optimises a risk-tolerance utility but does not quantify how aligned the resulting list is with the declared band in terms of ordinal distance. PC@ k is complementary: it is a metric for that alignment, and is computable on any ranker (with or without re-ranking).

2.4 Fairness-aware recommendation

Fairness-constrained recommender systems enforce group-level quality guarantees (e.g., equalised performance across demographic groups or item providers). PC@ k can be viewed through a similar lens: it measures per-group recommendation quality where groups are defined by declared MiFID risk bands, and PC-lift@ k normalises away the band-specific easiness of the asset universe, analogous to equalised-odds adjustments. The key difference is that PC@ k targets a regulatory suitability constraint rather than a social fairness objective.

All in all, existing FAR work optimises profitability or ranking, with risk-tolerance entering only as a post-hoc utility weight. No prior FAR evaluation metric operationalises MiFID II suitability as an ordinal coherence score over top- k lists. This paper closes that gap with PC@ k and a profile-coherent training-time objective.

3 Methodology

3.1 Profile Coherence

Ordinal band scale

Customers and assets share the same 4-point ordinal scale running from Conservative to Income to Balanced to Aggressive, coded 0 to 3. The customer band b_u is taken from FAR-Trans’s `riskLevel` field. FAR-Trans provides both declared values and regression-imputed `Predicted_*` entries, and both are mapped to the same ordinal scale. Customers whose entry is `Not_Available` or missing are excluded from all coherence computations. The asset band b_i follows the hierarchical rule in Section 4.2.

Pairwise discordance

$$d(u, i) = |b_u - b_i| \in \{0, 1, 2, 3\}. \quad (1)$$

A recommendation is *profile-coherent* iff $d \leq 1$. The tolerance of ± 1 band reflects MiFID II’s adjacency allowance for borderline suitability. Section 6 discusses this design choice further.

Aggregation

$$\text{PC@}k(u) = \frac{1}{k} |\{i \in \text{top}_k(u) : d(u, i) \leq 1\}|. \quad (2)$$

Two implementation rules complete the definition. First, recommenders returning fewer than k items use the actual number of items returned as the denominator. Second, customers with $b_u = \emptyset$ are dropped from the aggregate. Per-split PC@ k is the unweighted mean across eligible customers and the headline numerics average over all splits.

Band-conditional random baseline

The closed-form expectation of PC@ k under a uniformly random recommender for band b is given by

$$\pi(b) = \frac{|\{i \in A : |b - b_i| \leq 1\}|}{|A|}, \quad (3)$$

where A is the asset universe. Section 4.2 reports the FAR-Trans asset-band distribution and the resulting $\pi(b)$ values.

PC-lift@ k

Per-user PC@ k is normalised against the random reference at the customer’s own band:

$$\text{PC-lift@}k(u) = \frac{\text{PC@}k(u)}{\pi(b_u)}. \quad (4)$$

Lift > 1 beats the band-conditional random rate, while lift $= 1$ matches it. Because 77.9% of FAR-Trans assets sit in the Conservative, Income, and Balanced bands, a recommender that pushes Income assets at everyone would score well on raw PC@ k for any user whose tolerance set includes Income. Dividing by $\pi(b_u)$ strips the band-specific easiness of the asset universe and makes per-band values commensurate.

3.2 Stratified, Profile-Coherent LightGCN

Backbone

We choose LightGCN as the backbone because its differentiable loss and per-customer embeddings allow us to attach a coherence margin directly to the BPR objective. Random Forest is not extended because it ranks assets by predicted return from technical indicators alone and carries no customer-level signal to condition on.

Stratified architecture

We train four band-stratified sub-models, one per declared MiFID risk band. Each sub-model’s training graph contains only the interactions of customers assigned to that band. At inference, each customer is routed to the sub-model matching their declared band. Customers with no declared band default to the Balanced sub-model.

Profile-coherent margin loss

BPR trains the model to score an asset the customer actually bought (the positive item) higher than a randomly sampled unobserved asset (the negative item). Independently of this BPR sampling, we add a second margin term that samples one coherent asset i_c (satisfying $d(u, i_c) \leq 1$) and one discordant asset i_d (satisfying $d(u, i_d) > 1$) from the sub-model’s asset universe, and pushes the coherent score above the discordant score:

$$\mathcal{L}_{\text{coh}}(u) = \text{softplus}(s(u, i_d) - s(u, i_c)), \quad (5)$$

where $s(u, i)$ is the LightGCN inner-product score between customer u and asset i .

Total loss

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \omega \|\Theta\|_2^2 + \lambda \cdot \mathcal{L}_{\text{coh}}, \quad (6)$$

Here ω is the L2 weight decay on embedding parameters Θ and $\lambda \geq 0$ controls how strongly the coherence term influences training. BPR remains the primary ranking signal, and only the treatment trial activates the coherence term.

Ablation Setup

Setting $\lambda = 0$ trains stratified BPR without the coherence term, isolating the effect of the stratified architecture. Setting $\lambda = 1$ adds the coherence margin on top. Comparing the two trials lets us attribute gains separately to stratification and to the PC-loss.

4 Experimental Setup

4.1 Dataset

In addition to the Buy transactions and risk-band fields described in Sections 1 and 3, FAR-Trans [Sanz-Cruzado *et al.*, 2024] provides 159,136 Sell transactions and daily closing prices spanning January 2018 to November 2022. Of the 29,090 retail customers, 28,770 (99.0%) carry a declared or regression-imputed MiFID risk band; the remaining 320 are excluded from all coherence computations. Customer profiles also include `customerType` and `investmentCapacity`, both of which enter the RQ2 regression controls in Section 4.5.

4.2 Asset Risk Band Assignment

Each asset is assigned to one of the four MiFID risk bands by a three-step hierarchical rule, applied in order.

Step 1: subcategory metadata

Mutual funds are labelled directly from their subcategory: Money Market funds are treated as Conservative, Bonds as Income, Balanced as Balanced, and Equity or Large Cap as Aggressive. Bonds are labelled by issuer type: Government as Conservative, Corporate as Income, and any remaining bond subcategory as Income by default.

Parameter	Search space	Best	Role
LightGCN (tuned on average nDCG@10)			
<code>embedding_dim</code>	{64, 128}	64	grid
<code>num_layers</code>	{2, 3}	3	grid
<code>learning_rate</code>	$\{10^{-2}, 10^{-3}\}$	10^{-3}	grid
<code>weight_decay</code>	10^{-5}	—	fixed
<code>keep_prob</code>	0.6	—	fixed
<code>num_epochs</code>	50	—	fixed
<code>batch_size</code>	1024	—	fixed
Random Forest (tuned on average ROI@10)			
<code>num_estimators</code>	{20, 30, 40, 50}	40	grid
<code>max_depth</code>	{15, 25, 50}	50	grid
<code>horizon</code>	6 months	—	fixed
<code>random_state</code>	42	—	fixed

Table 1: Baseline hyperparameter grids and best-trial values. “grid” rows are searched, “fixed” rows are held constant.

Step 2: volatility quartiles for stocks

Stocks carry no subcategory tag we can use, so they fall through to a volatility rule. For each ISIN we compute the annualised standard deviation of daily log returns on a trailing 252-day window, then bucket all stocks into the four bands by the stock-only quartile thresholds (q_1, q_2, q_3) .

Step 3: residual default

Any asset that falls through both prior steps is assigned to Balanced.

Resulting distribution. Applying the three steps above produces an asset-band distribution of (190, 333, 105, 178) across (Conservative, Income, Balanced, Aggressive) over $|A| = 806$ assets. Under the formula defined in Section 3.1, this yields a band-conditional random baseline of $\pi(b) = (0.65, 0.78, 0.76, 0.35)$.

4.3 Temporal Split Protocol

Evaluation uses 69 temporal splits on an evenly spaced trading-day grid between August 2019 and May 2022, yielding approximately biweekly split points. For each split, the training set contains all Buys in $[0, \text{time_point}]$ and the test set contains all Buys in $[\text{time_point}, \text{test_end}]$, deduplicated against training and filtered to the training-asset universe. Eligible customers have at least one interaction in both training and test, while eligible assets have prices on both endpoints. In RQ4 (Section 5.4), per-split metric deltas are paired against the global LightGCN baseline to quantify the effect of each intervention component.

4.4 Baselines and Hyperparameter Grids

Table 1 reports the search grids, fixed values, and best-trial selections for the two FAR-Trans baselines. Both grids are swept with Ray Tune, with LightGCN tuned on average nDCG@10 and Random Forest on average ROI@10 across the 69 splits.

The stratified PC-LightGCN inherits the best LightGCN trial’s backbone hyperparameters. Two trials are evaluated, $\lambda = 0$ (BPR only, isolating the stratified architecture) and $\lambda = 1$ (coherence margin activated), on the same 69 splits.

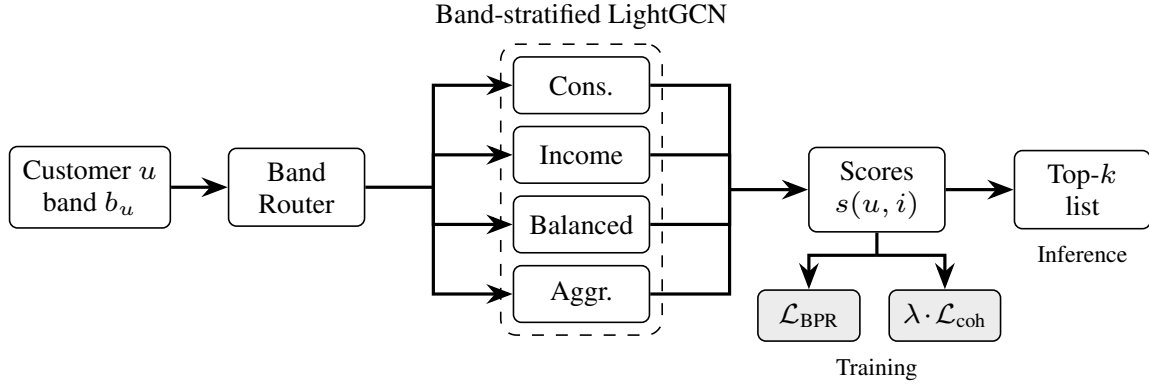


Figure 1: PC-LightGCN architecture. Each customer is routed by declared MiFID band to a dedicated sub-model. Training combines $\mathcal{L}_{\text{BPR}}$ with a coherence margin $\lambda \cdot \mathcal{L}_{\text{coh}}$. At inference the sub-model produces the top- k list directly.

4.5 Statistical Inference

Three of the four research questions from Section 1 require statistical analysis beyond the headline metrics. RQ1 is descriptive and needs no inference test. The remaining three map to the following procedures.

- **RQ2: Does discordance carry a return penalty?** We fit a transaction-level OLS (Equation 7) on 208,029 priced Buys with cluster-robust standard errors on `customerID` and a two-sided Wald test at $\alpha = 0.05$. Clustering is warranted because each customer contributes roughly 8 Buys on average, so within-customer residuals share unobserved shocks.
- **RQ3: Do baselines lift coherence on every band?** For each (declared band, model) cell, we average per-customer PC@10 unweighted over all (customer, split) pairs. Per-band lift is then the ratio of that cell mean to the random baseline $\pi(b)$, which normalises away the band-specific easiness of the asset universe (Section 3.1).
- **RQ4: Does PC-LightGCN improve coherence, and at what cost?** We average each metric over the 69 splits and compute paired per-split deltas against the baseline LightGCN. The win rate reports the fraction of splits on which the treatment outperforms the comparator.

The RQ2 OLS specification is given below and interpreted in Section 5.2:

$$\begin{aligned}
 r = & \beta_0 + \beta_1 \cdot \text{is_coherent} \\
 & + \beta_2 \cdot \text{asset_volatility} \\
 & + \mathbf{C}(\text{type})\gamma \\
 & + \mathbf{C}(\text{year})\delta \\
 & + \varepsilon.
 \end{aligned} \tag{7}$$

where $r = (p_{\text{end}} - p_{\text{start}})/p_{\text{start}}$ is the 6-month realised return as a decimal fraction (e.g., $r = 0.03$ means a 3 percentage-point gain) and `is_coherent` is binary ($d \leq 1$). Prices are matched via a backward-nearest join: the start price is the most recent close on or before the trade date, and the end price is the most recent close on or before the date six months later. Transactions without a price match within 7 days at either endpoint are dropped.

4.6 Reproducibility

All code, configuration grids, and evaluation artefacts are available in the project repository at <https://github.com/Samthesimpsons/SMU-Capstone>.

5 Results and Analysis

We present the four research questions as a diagnostic-to-remedy chain. RQ1 establishes that profile-discordance is prevalent and structural, RQ2 shows that it carries a measurable return penalty, RQ3 audits where the existing baselines sit on the coherence axis, and RQ4 demonstrates a training-time remedy that closes the gap.

5.1 RQ1: Profile-Discordance is Prevalent and Structural

We characterise discordance in four slices, starting from the pooled transaction rate and drilling into the customer, band, and year dimensions in turn. At the transaction level (Figure 2), of 228,241 scoreable Buys:

1. 18.6% violate the declared band by two or more steps.
2. 81.4% are coherent, of which 34.1% strictly ($d = 0$).
3. Mean discordance is 0.874 bands across all scoreable Buys, indicating that when mismatches occur they are usually a single band step.

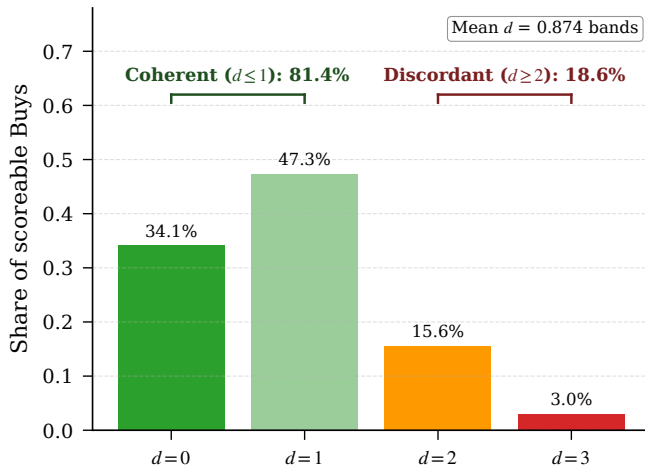


Figure 2: Distribution of pairwise discordance d over 228,241 scoreable Buys. Most mass sits at $d \leq 1$ (coherent), with non-negligible density at $d \geq 2$ (discordant).

The pooled rate cannot tell us whether these violations are i.i.d. transaction-level slips spread thinly across the customer base or a structural trait concentrated in a subset. Slicing by customer settles the question. The per-customer discordant share (Figure 3) is sharply bimodal, concentrating in the lowest and highest bins rather than clustering near the 18.6% population mean. More explicitly, 64.4% of banded customers are fully coherent across every Buy and 17.2% are fully discordant. Hartigan's dip test on the 28,770 per-customer values rejects unimodality ($D = 0.086$, $p < 10^{-15}$), confirming that discordance is a persistent customer-level trait rather than i.i.d. transaction-level noise.

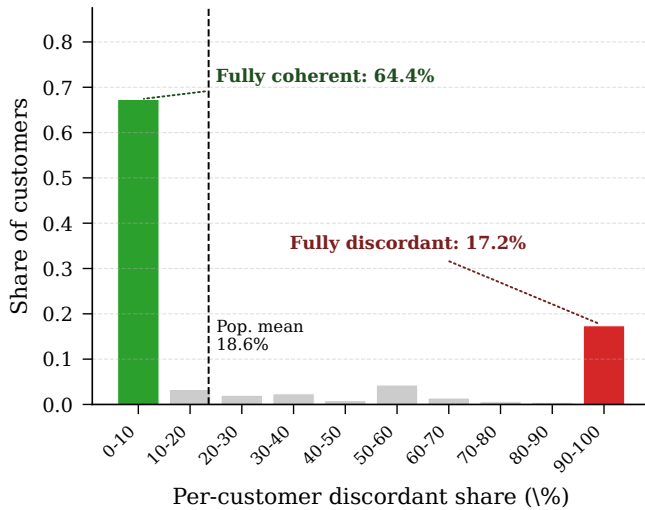


Figure 3: Per-customer share of discordant Buys is sharply bimodal, concentrating in the lowest and highest bins and ruling out i.i.d. transaction-level noise.

Since the trait is customer-level, the next question is which customers carry it. Coherence per declared band is U-shaped (Figure 4). The two centre bands sit at $\sim 90\%$, while Conser-

vative drops to 45.1% (possibly reflecting a “reach for yield” tendency) and Aggressive to 56.2% (possibly reflecting a regression toward safer assets). This concentration of mismatch at the ordinal extremes corroborates the bimodality finding from the customer slice.

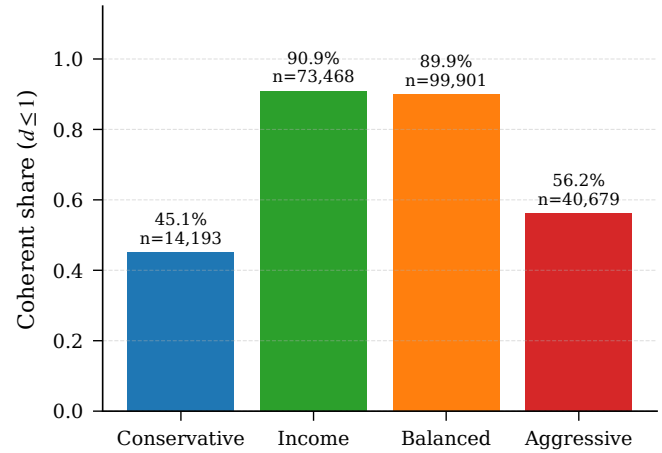


Figure 4: Coherence share by declared band. The U-shape supports the bimodality result of Figure 3: discordance concentrates on the ordinal extremes.

A final slice rules out a regime-driven artefact. Mean discordance is flat across 2019 to 2022 (within 0.02 bands, see Figure 5), so the pattern is not the product of any single year's market conditions. The 2018 value (0.98) sits higher but covers a small slice. MiFID II came into force on 2018-01-03, the exact start of FAR-Trans, so we hypothesise the elevation as a transition effect. However, this hypothesis is not tested and is potential future work.

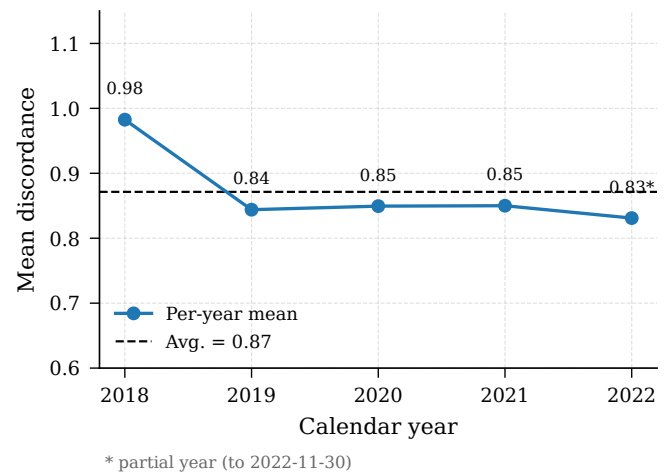


Figure 5: Mean discordance by calendar year. The 2019 to 2022 cluster sits within 0.02 bands, with 2018 elevated (transition-effect hypothesis).

5.2 RQ2: Coherent Buys Earn a Conditional Return Premium

Having established that discordance is a persistent customer-level trait concentrated at the ordinal extremes, the natural follow-up is whether it matters economically: do discordant Buys actually earn less?

RQ2 asks whether profile-coherent Buys earn different realised returns from discordant ones, after controlling for asset volatility, customer segment, and calendar year. We test this via the OLS specification in Equation 7, where the coefficient of interest is β_1 on `is_coherent`. The null hypothesis $H_0: \beta_1 = 0$ states that coherence has no conditional effect on 6-month returns, and we evaluate it with a two-sided Wald test at $\alpha = 0.05$ using cluster-robust standard errors on `customerID`. The model is fit on 208,029 priced Buys, and Table 2 reports the full coefficient table.

Coherent Buys earn +2.94 percentage points (pp) higher 6-month realised return than discordant ones, conditional on volatility, segment, and year. The positive sign with $p < 10^{-12}$ provides strong evidence against the prior that regulatory coherence imposes a return tax. The raw slice gap is +3.31 pp, so controls absorb only ~0.4 pp (~11% of the unconditional gap). This implies that coherence is not a volatility, segment, or year confound. However, it is important to note that unobserved confounders such as financial literacy, advisor channel, and account constraints are not ruled out. The model’s adjusted R^2 is 0.051, meaning 95% of return variance is driven by factors outside the specification; this is typical of cross-sectional return regressions where idiosyncratic price movements dominate. The low R^2 does not affect coefficient inference: valid hypothesis testing requires correctly specified standard errors, not high explanatory power, and the cluster-robust SEs account for within-customer correlation across the 27,873 customer clusters.

The volatility coefficient is -10.3 pp per unit, consistent with low-volatility outperformance over 2019 to 2022 and the 2022 rate shock that hit high-beta names. All four year dummies reject H_0 , reflecting distinct market regimes relative to the 2018 reference year. The 2019 coefficient captures the equity recovery from the late-2018 selloff, 2020 is the strongest at +21.1 pp driven by the post-COVID V-shaped rebound, and 2021 reflects continued recovery under accommodative monetary policy before late-2021 rate-hike expectations set in. The 2022 coefficient turns negative at -2.0 pp as aggressive Fed hikes triggered a broad fixed-income drawdown. All four `customerType` dummies fail to reject H_0 , indicating that segmentation contributes no detectable return signal. Together with the bimodality found in RQ1, this suggests that customer type does not capture the persistent discordance trait either. The coherence premium therefore survives all controls we can measure, motivating the question of whether existing recommenders produce coherent lists at all.

5.3 RQ3: Both Baselines Under-Serve at Least One Band

With a confirmed return premium for coherent Buys, we now audit whether the two FAR-Trans baselines actually deliver profile-coherent recommendations across all four risk bands.

RQ3 asks whether the two FAR-Trans baselines produce profile-coherent recommendations across all four risk bands. We begin by comparing their aggregate performance at primary-metric optima, summarised in Table 3. LightGCN wins on ranking quality with `nDCG@10` of 0.330, `Recall@10` of 0.495, and `PC@10` of 0.794, but loses on `ROI@10` at -0.5%/mo. Conversely, Random Forest wins on `ROI@10` at +1.4%/mo but ranks near random on `nDCG@10` of 0.019 and `Recall@10` of 0.036. Both appear adequate on aggregate `PC@10`, with LightGCN at $1.17\times$ lift and Random Forest at $1.06\times$, but the aggregate hides per-band failures.

However, aggregate `PC@10` averages across customer bands whose random baselines $\pi(b)$ vary from 0.35 to 0.78, so a single number can mask which bands are actually being served. Table 4 decomposes `PC@10` by declared band to expose these differences. Per-band lift moves $0.75 \rightarrow 1.13 \rightarrow 1.17 \rightarrow 1.35$ for LightGCN and $0.52 \rightarrow 0.70 \rightarrow 1.03 \rightarrow 1.88$ for Random Forest across Conservative $\rightarrow$ Income $\rightarrow$ Balanced $\rightarrow$ Aggressive. For both, lift rises monotonically with the declared band: recommended assets skew higher-risk. LightGCN’s range is flatter, which we attribute to its per-user embeddings shifting each customer’s top- k partly toward their tolerance window. Recall that Random Forest carries no customer signal.

Band	Model	N	$\pi(b)$	PC@10	PC-lift@10
Conservative	RF	5,348	0.65	0.34	0.52
Conservative	LightGCN	5,348	0.65	0.48	0.75
Income	RF	45,142	0.78	0.54	0.70
Income	LightGCN	45,142	0.78	0.88	1.13
Balanced	RF	61,181	0.76	0.79	1.03
Balanced	LightGCN	61,181	0.76	0.89	1.17
Aggressive	RF	25,367	0.35	0.66	1.88
Aggressive	LightGCN	25,367	0.35	0.47	1.35

Table 4: Per-band `PC@10` and `PC-lift@10` versus the band-conditional random baseline $\pi(b)$ for the two FAR-Trans baselines at their primary-metric optima, aggregated over 69 splits. `PC-lift@10` = `PC@10` / $\pi(b)$.

Measured against the full FAR-Trans customer base (28,770 banded customers), LightGCN under-serves the 19.7% Conservative segment while Random Forest under-serves 60.9% (Conservative and Income combined). Aggressive’s high lift is partly mechanical: $\pi(\text{Aggressive}) = 0.35$ is the lowest band, so any model skewing higher-volatility clears that bar easily. Neither baseline has band labels or a band-aware loss, so per-band coherence is a passive byproduct of training rather than a controlled output.

5.4 RQ4: Stratification + PC-Loss Closes the Per-Band Deficit

RQ3 showed that both baselines under-serve at least one risk band, with the deficit concentrated on the Conservative segment for LightGCN and on both Conservative and Income for Random Forest. RQ4 asks whether a band-aware architecture can close this gap without sacrificing ranking quality or return. PC-LightGCN (Section 3.2) tackles this by routing each customer to a band-stratified sub-model and adding a

Term	Estimate	Std. Error	95% CI	<i>p</i>
Intercept	−0.0471	0.0141	[−0.0746, −0.0195]	8.1×10^{-4}
is_coherent	+0.0294	0.0040	[+0.0215, +0.0372]	2.3×10^{-13}
asset_volatility	−0.1025	0.0165	[−0.1349, −0.0701]	5.5×10^{-10}
C(customer_type) [Legal Entity]	+0.0154	0.0486	[−0.0798, +0.1106]	0.751
C(customer_type) [Mass]	+0.0011	0.0132	[−0.0248, +0.0269]	0.936
C(customer_type) [Premium]	+0.0058	0.0135	[−0.0206, +0.0322]	0.667
C(customer_type) [Professional]	+0.0247	0.0169	[−0.0084, +0.0578]	0.143
C(year) [2019]	+0.1759	0.0129	[+0.1507, +0.2012]	1.4×10^{-42}
C(year) [2020]	+0.2107	0.0061	[+0.1986, +0.2227]	6.2×10^{-259}
C(year) [2021]	+0.0754	0.0053	[+0.0651, +0.0858]	4.1×10^{-46}
C(year) [2022]	−0.0195	0.0045	[−0.0283, −0.0107]	1.4×10^{-5}

Table 2: Full RQ2 OLS coefficient table on 208,029 priced Buys, cluster-robust SE on customerID. Reference cell: Inactive customerType (the alphabetically first segment in FAR-Trans), year 2018, asset_volatility = 0, is_coherent = 0.

Model	nDCG@10	ROI@10	Recall@10	PC@10	PC-lift@10
LightGCN	0.330	−0.005	0.495	0.794	1.17×
RF	0.019	+0.014	0.036	0.667	1.06×

Table 3: RQ3 aggregate metrics at primary-metric optima, averaged over 69 splits. ROI@10 is per-month. PC-lift@10 is PC@10 divided by the band-conditional random baseline $\pi(b)$.

coherence margin to the BPR objective. We evaluate two ablations against the baseline LightGCN: $\lambda = 0$, which isolates the effect of stratification alone, and $\lambda = 1$, which activates the coherence margin on top of stratification.

Table 6 reports aggregate metrics for the three trials averaged over 69 splits. Figure 6 complements the table by showing the per-split win rate of each ablation against the baseline, defined as the fraction of the 69 splits on which the trial outperforms the comparator (Section 4.5). This indicates how consistently each trial wins rather than whether a single strong split drives the aggregate. Figure 7 goes one step further, plotting the paired per-split deltas so the magnitude and direction of each metric shift are visible split by split.

The headline result is that the $\lambda = 1$ treatment lifts PC@10 from 0.794 to 0.918, a +12.42 pp gain, and PC-lift from 1.17× to 1.40×. This improvement is consistent across all temporal splits: Figure 6 shows a 100% win rate on PC@10 across all 69 splits, ruling out the possibility that a few strong splits drive the aggregate.

Because the treatment bundles two architectural changes, the $\lambda = 0$ ablation lets us isolate how much each piece contributes to the +12.42 pp PC@10 gain. Stratification alone supplies +2.75 pp of PC@10, about a quarter of the full gain. Importantly, it does so at virtually no cost to the other metrics. The nDCG@10 delta is +0.0022 with a 56.5% win rate, Recall@10 shifts by +0.0063 with a 72.5% win rate, and ROI@10 moves by −0.0019 pp/mo with a 27.5% win rate. All three deltas are small enough that none registers as a meaningful effect, making stratification the cheap component of the intervention.

Adding the coherence margin on top buys the remaining +9.68 pp of PC@10, roughly four-fifths of the total, winning on every split. This larger gain comes at a measurable but bounded ranking cost of −2.23 pp of nDCG@10 and −3.38

pp of Recall@10, both lost on every split.

Aggregate ROI@10 is essentially flat across all three trials. The stratification ablation shifts ROI@10 by −0.0019 pp/mo with a 27.5% win rate, while the full treatment shifts it by −0.0018 pp/mo with a 33.3% win rate. The coherence margin step on its own contributes a near-zero +0.0002 pp/mo with a 58.0% win rate. The small ROI cost therefore lives in the stratification step rather than the coherence margin, and is too small to distinguish from zero in Table 6.

Together with the coherence premium established in RQ2, this null aggregate-ROI effect indicates that the recommender can be made meaningfully more profile-coherent without giving up the return story.

Table 5 decomposes PC-lift@10 by declared band to trace how the intervention resolves the per-band deficit from RQ3. Conservative, which sat at 0.75× under the baseline, climbs to 0.91× under stratification alone but remains below the random baseline; only after adding the coherence margin does it cross to 1.19×. Aggressive exhibits a non-monotonic pattern: stratification alone reduces lift from 1.35× to 1.17×, likely because the smaller Aggressive training graph yields lower-quality embeddings in isolation, but the coherence margin more than recovers the loss at 1.97×. Income and Balanced, already above-random under the baseline, improve modestly without being sacrificed. Under $\lambda = 1$, all four bands clear the random baseline for the first time.

Band	Baseline	$\lambda = 0$	$\lambda = 1$
Conservative	$0.75\times$	$0.91\times$	$1.19\times$
Income	$1.13\times$	$1.20\times$	$1.26\times$
Balanced	$1.17\times$	$1.22\times$	$1.28\times$
Aggressive	$1.35\times$	$1.17\times$	$1.97\times$

Table 5: Per-band PC-lift@10 for the three RQ4 trials. Values above $1.00\times$ beat the band-conditional random baseline $\pi(b)$.

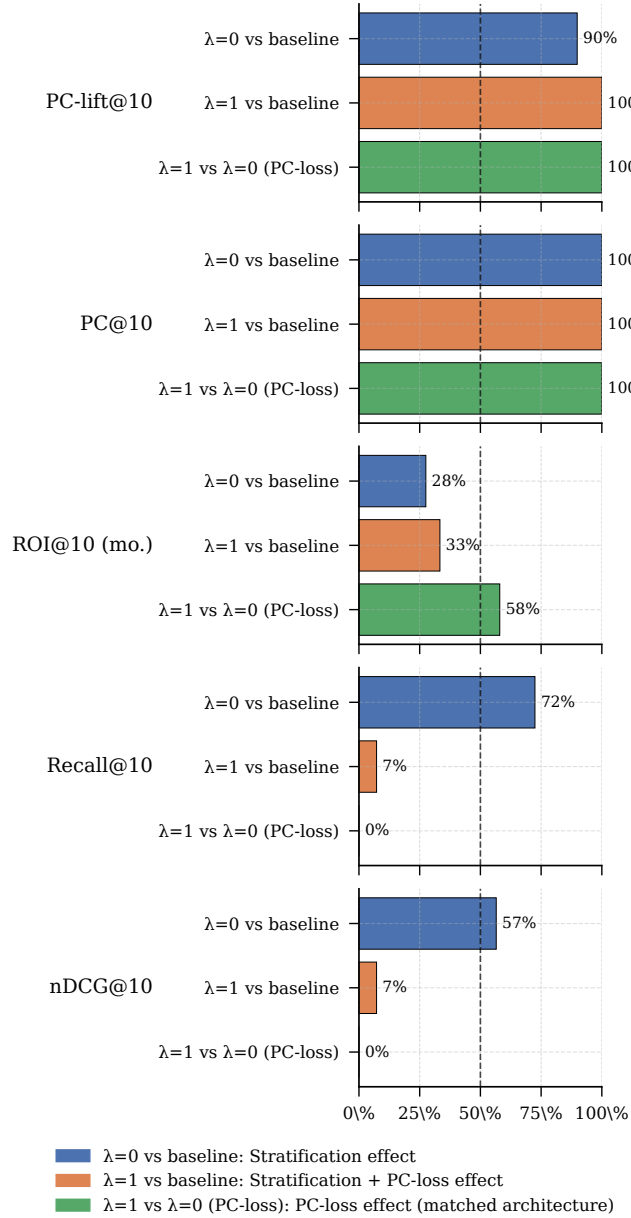


Figure 6: Per-split win rates of the $\lambda = 0$ and $\lambda = 1$ trials against the global LightGCN baseline.

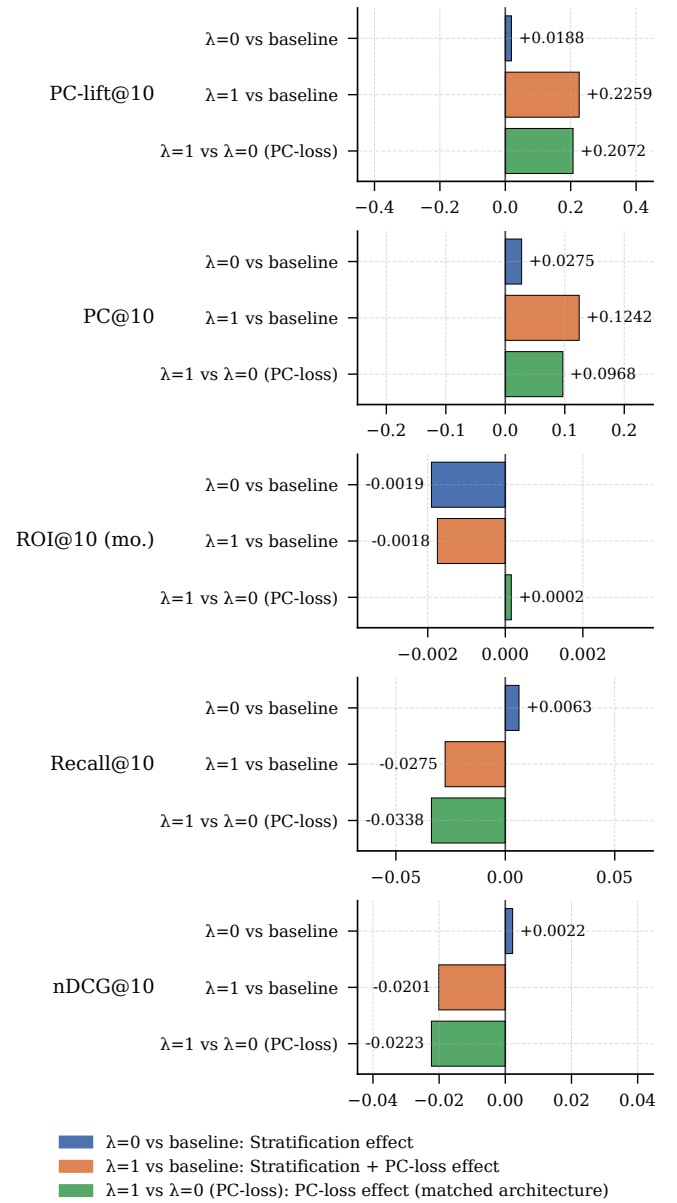


Figure 7: Paired per-split deltas of $\lambda = 1$ vs baseline LightGCN. PC@10 is positive on every one of 69 splits.

6 Limitations and Future Work

We acknowledge several boundaries of the current study, each of which opens a concrete direction for future work.

- *Observational nature.* The RQ2 coherence premium is associational as unobserved confounders such as financial literacy, advisor channel, account constraints, and prior portfolio composition are not ruled out by the volatility, segment, or year controls. Similarly, the elevated 2018 mean discordance is hypothesised as a MiFID II transition effect but not directly tested against a pre-MiFID counterfactual. A natural next step would be to apply causal inference techniques such as instrumental variables or difference-in-differences on datasets that span a

Trial	nDCG@10	ROI@10	Recall@10	PC@10	PC-lift@10
Baseline LightGCN	0.330	-0.005	0.495	0.794	1.17×
$\lambda = 0$	0.332	-0.007	0.501	0.821	1.19×
$\lambda = 1$	0.309	-0.007	0.467	0.918	1.40×

Table 6: RQ4 aggregate metrics, averaged over 69 splits. ROI@10 is per-month. PC-lift@10 is PC@10 divided by the band-conditional random baseline $\pi(b)$, expressed as a multiple.

regulatory boundary.

- *Metric and band-assignment design choices.* The tolerance $d \leq 1$ is set rather than derived, binarising a 4-point ordinal distance, and the asset-band assignment in Section 4.2 is a hierarchical heuristic. Future work could explore data-driven tolerance calibration and validate the band assignment against external MiFID II classifications from fund providers.
- *Intervention scope and tuning.* This study extends only LightGCN, leaving sequential, transformer, and content-based recommenders as candidates for the same stratification and coherence-margin treatment. Within LightGCN, $\lambda = 1$ is one point on a Pareto frontier between PC and ranking-quality metrics, and the intervention inherits the RQ3 best-trial backbone hyperparameters rather than being re-tuned end-to-end. The reported coherence gains are therefore a lower bound on what a full frontier sweep or end-to-end re-tuning could achieve, and the reported nDCG and Recall costs are correspondingly an upper bound.
- *Generalisation and measurement scope.* Results are demonstrated on FAR-Trans under MiFID II’s 4-band scale. Extending the evaluation to other jurisdictions, longer time horizons, or alternative asset universes would strengthen the generalisability of PC@ k as a suitability metric. Additionally, RQ2 measures realised return on executed Buys while RQ4 measures forward return on top- k recommended slates that may never be acted on. Bridging this gap through online evaluation or counterfactual off-policy estimation is a promising direction.

7 Conclusion

The central finding of this paper is that the regulatory objective (MiFID-aligned band coherence) and the economic objective (realised return) are not in structural conflict on FAR-Trans: a recommender can be made meaningfully more profile-coherent while keeping the return story the data already supports. This reframes coherence as an architectural choice rather than a tradeoff to be managed post-hoc, treating compliance as a training-time objective expressible as an ordinal distance on a customer-asset metric pair.

The four research questions assemble a diagnostic-to-remedy chain that supports this conclusion:

- **RQ1** establishes that the diagnostic axis is real and structural at the customer level, not transaction-level noise.
- **RQ2** removes the economic objection to optimising for coherence: coherent Buys earn a statistically significant return premium over discordant ones, providing evidence

against the hypothesis that regulatory alignment imposes a return tax.

- **RQ3** locates the existing FAR-Trans baselines on the coherence axis and shows that both under-serve specific declared bands in ways the aggregate metrics hide.
- **RQ4** demonstrates that a stratified, profile-coherent LightGCN extension closes the per-band coverage deficit at no aggregate ROI cost, with the gain decomposing cleanly into a free stratification component and a bounded-cost PC-loss component.

More broadly, the approach illustrates that regulatory suitability constraints can be encoded as training-time objectives rather than handled through post-hoc filtering, opening a path for recommender systems in other regulated domains (e.g., insurance product suitability, credit risk appetite matching) to embed compliance directly into their optimisation loop.

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